# Youngchan Kim

Ph.D. student @ MPI for Informatics

*kyc9618@gmail.com* | *https://youngchan-k.github.io*

## Education

**Max Planck Institute for Informatics / Saarland University** — *Apr 2026 – Present*
Ph.D. in Computer Science — *Saarbrücken, Germany*

**POSTECH** — *Mar 2022 – Feb 2024*
M.S. in Artificial Intelligence — *Pohang, South Korea*

**Yonsei University** — *Mar 2016 – Feb 2022*
B.S. in Mathematics & Electrical and Electronic Engineering — *Seoul, South Korea*

## Research Experience

**Visual Computing and AI Department, MPI for Informatics** — *Apr 2026 – Present*
Doctoral Researcher — *Saarbrücken, Germany*

- Advisor: Prof. Christian Theobalt
- (1) Human Performance Capture / (2) Human-Object Interaction / (3) Relightable Human Avatars

**Computer Graphics Lab, POSTECH** — *Mar 2022 – Feb 2024*
Master's student — *Pohang, South Korea*

- Advisor: Prof. Seung-Hwan Baek
- (1) High-dimensional Light Properties / (2) Light-Matter Interaction Modeling

## Publications

### * Multi-Dimensional Light Properties *

[1] **Hyperspectral Polarimetric BRDFs of Real-world Materials**
Yunseong Moon, Ryota Maeda, Suhyun Shin, Inseung Hwang, **Youngchan Kim**, Min H. Kim, Seung-Hwan Baek
*SIGGRAPH Asia 2025*

[2] **Spectral and Polarization Vision: Spectro-polarimetric Real-world Dataset**
Yujin Jeon*, Eunsue Choi*, **Youngchan Kim**, Yunseong Moon, Khalid Omer, Felix Heide, Seung-Hwan Baek
*CVPR 2024 (highlight)*

[3] **Neural Spectro-polarimetric Fields**
**Youngchan Kim**, Wonjoon Jin, Sunghyun Cho, Seung-Hwan Baek
*SIGGRAPH Asia 2023*

## Work Experience

**NationA** *Aug 2025 – Nov 2025*
AI/ML Engineer *Seoul, South Korea*

- Reproduce AI character motion generation algorithms for realistic and temporally coherent results
- Deploy motion synthesis and editing models via APIs to support 3D animation and content creation

**DEEPX** *May 2025 – Jul 2025*
AI Compiler Engineer *Seongnam, South Korea*

- Restructure threading and tiling execution to enhance readability and support more flexible patterns
- Design and implement a layer numbering strategy for ONNX to optimize computation efficiency in NPU

**Beeble** *Apr 2024 – May 2025*
AI Researcher *Seoul, South Korea*

- Develop human-centric vision models including PBR map extraction and monocular depth estimation
- Build a virtual light stage for multi-view 3D human rendering with PBR maps and controllable lighting
- Optimize AI models to extract pixel-accurate VFX passes (albedo, normal, depth, etc.) from video inputs
- Implement real-time video relighting with extracted PBR maps and HDRI light generation in web editor

## Teaching Experience

### Lecturer

- Data Analyst Bootcamp (Cohort 12) - Codeit *Feb 2026 – Apr 2026*
- Vibe Coding with Antigravity - Yeungnam University *Dec 2025*
- AI Integration for Digital Transformation in Practice - KOSME *Nov 2025*
- Python and Data Analytics for Digital Transformation - Samsung Electronics *Apr 2024*

### Teaching Assistant

- AI Engineer Bootcamp (Cohort 9) - Codeit *Dec 2025 – Apr 2026*
- Deep Learning and Computer Vision - POSCO Group *Oct 2022 / Jul 2023*
- CSED700G TinCS: Computational Imaging - POSTECH *Spring 2023*

## Academic Activities

**Reviewer** Pacific Graphics 2023

## Skills

**Languages** Korean (Native - 5/5), English (Conversational - 3~4/5)

**Programming & SW** Python, PyTorch, TensorFlow, OpenCV, NumPy, Matplotlib, C/C++, Linux, Docker



Last updated on March 14, 2026

**Graphics & Rendering** Blender, Google Filament, Three.js, Mitsuba 2/3

## Reference

### Prof. Christian Theobalt

Ph.D. advisor

- Director, Visual Computing and AI Department, MPI for Informatics
- Professor, Department of Computer Science, Saarland University
- Director, Saarbrücken Center for Visual Computing, Interaction and Artificial Intelligence
- *theobalt@mpi-inf.mpg.de* | *https://people.mpi-inf.mpg.de/~theobalt*

### Dr. Rishabh Dabral

Ph.D. advisor

- Research Group Leader, 3D Visual Intelligence Research Group, MPI for Informatics
- *rdabral@mpi-inf.mpg.de* | *https://rishabhdabral.github.io*

### Prof. Seung-Hwan Baek

M.S. advisor

- Associate Professor, Department of Computer Science and Engineering, POSTECH
- Affiliated Professor, Graduate School of Artificial Intelligence, POSTECH
- *shwbaek@postech.ac.kr* | *https://www.shbaek.com*

### Prof. Ki Jun Yu

B.S. advisor

- Associate Professor, Department of Electrical and Electronic Engineering, Yonsei University
- *kijunyu@yonsei.ac.kr* | *https://bioelectronics.yonsei.ac.kr/*